ANWE - Design for Values Assessment

Animal Welfare and Ecosystem Restoration

Sierra de la Culebra, Zamora, Spain / June 2022 wildfire

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Review status: Unreviewed draft. Qualified expert identity and endorsement are not verified.

1. Executive Summary

Environmental evidence

The calculated regional mean dNBR (satellite change indicator) is 0.114. It describes spectral change across the analysed region, including unburned land; it is not a direct damage measure.

Biodiversity evidence

Inside the mapped fire footprint, the eligible retrieved sample contains 13 before-window records and 0 after-window records. Many records may have uncertain locations. These are observation counts, not animal populations.

Options evaluated

Natural regeneration, assisted ecological restoration and active habitat restoration. These are conditional options for investigation, not an automatically selected recommendation.

Main uncertainty

Current animal use, habitat function and intervention effects have not been field-verified. Observation effort and location precision limit biodiversity comparisons.

Decision status

No option is authorised or implementation-ready. Human ecological review, landowner consent and applicable permissions remain necessary.

AI participation: An actual model draft is included; it is not an expert-approved finding.

Reading path: 2. Wildfire/environment → 3. Biodiversity → 4. Options → 5. Values → 6. Next steps. Methods, species lists, licences and glossary follow in appendices.

2. Wildfire Context and Environmental Evidence

Fires were reported from 2022-06-15. The extinguishment date is unverified. The mapped observation dates from 25 June 2022; it is not an exact final incident perimeter.



Rust outlines: Copernicus Emergency Management Service (CEMS), activation EMSR580. (c) European Union. Generalised source geometry, with an additional 120 m simplification for this overview. UTM 29N grid; no animal locations or basemap.

Actual image acquisitions: before 2022-06-10T11:06:19.024Z; later 2022-07-05T11:06:31.024Z. The later image describes that date, not automatically an immediate post-fire condition.

Indicator	Regional mean	What it means
Pre-fire NDVI	0.482	Average vegetation greenness in the earlier image.
Later-image NDVI	0.385	Average greenness in the selected later image.
dNBR	0.114	Difference in the burn-sensitive spectral index between images.

What does this tell us?

NDVI means Normalized Difference Vegetation Index; NBR means Normalized Burn Ratio; dNBR is before NBR minus later NBR. They support investigation of vegetation/spectral change, not percentages of damage, animal deaths, soil moisture or proven habitat recovery. Regional means include unburned land. Methods and full precision: Appendix A.

European Forest Fire Information System (EFFIS) historical severity data are unavailable here. Later observations may also reflect the separate July 2022 fire.

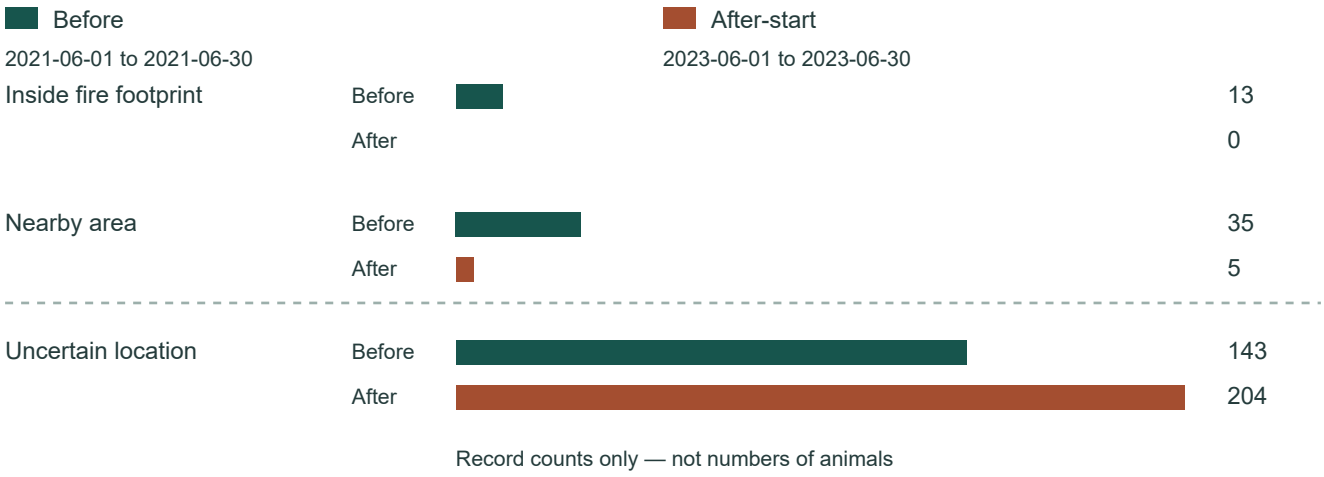
3. Biodiversity Before and After the Fire

Before: 2021-06-01 to 2021-06-30. After-start: 2023-06-01 to 2023-06-30. Group: All; outside buffer: 5 km.

These windows match calendar-season boundaries to reduce seasonal differences; sampling effort may still differ.

The same June 2022 outline is used for both periods. Before records describe places later mapped as burned. The nearby buffer extends outward from the boundary and is not a validated unaffected control. Uncertain location is a data-quality category, not a third geographical area.

Location category	Before records	Before species	After records	After species
Inside mapped fire footprint	13	6	0	0
Nearby comparison area	35	16	5	4
Uncertain location	143	62	204	79



Record counts only — not numbers of animals

What the data shows

The table counts date-eligible retrieved occurrences and distinct identified species. Spatially uncertain records are separate from confident inside/nearby assignments. Complete species sets and quality details are in Appendix B.

What the data does not prove

Fewer records do not establish disappearance, population decline or reduced biodiversity. More records do not establish recovery. GBIF is an index of reported observations, not a census.

What additional evidence is needed

Use comparable field surveys and information on observer effort, reporting, date precision and habitat use. Interpret later records with the July-fire confounder in mind.

4. Restoration Strategy Comparison

These authored options identify possibilities to investigate. Benefits and harms are conditional, not measured local effects or model-generated findings. No option is labelled recommended.

Compare	Natural regeneration	Assisted restoration	Active habitat restoration
What it involves	Protect existing recovery processes and monitor vegetation and wildlife. Avoid soil disturbance; no routine planting, feeding or water provisioning.	Consider targeted erosion control and habitat protection only where field evidence supports them. Native planting is one possible measure if regeneration is inadequate; it is not a default.	Evaluate more extensive habitat work only if surveys demonstrate a need. Compare measures against natural recovery; agree timing, monitoring and stopping criteria with qualified ecologists.
Potential benefits	May retain locally adapted vegetation and surviving refugia May avoid machinery disturbance to animals and soil organisms	Targeted erosion control may protect aquatic habitat Protecting refuge patches may preserve shelter and movement routes	May restore connectivity where severe fragmentation is demonstrated Could address persistent habitat loss when lighter measures are insufficient
Possible harms / ecological limitations	May leave erosion or habitat loss unresolved where recovery is insufficient Acute welfare concerns may require a separate qualified assessment	Ground works can harm soil biota and nesting animals Planting may fail under drought; barriers may obstruct wildlife	Machinery and clearing may damage refugia and less visible animals Species introductions, feeding or water provisioning create separate disease and welfare risks and are not prescribed here
When inappropriate	May be inappropriate where field evidence shows severe erosion, failed regeneration or invasive dominance.	May be inappropriate where recovery is already adequate or disturbance costs exceed expected benefits.	May be inappropriate without evidence that lighter interventions are insufficient or where sensitive breeding habitat would be disturbed.
Critical field focus	Verify whether natural regeneration is sufficient and erosion can be managed without intrusive work.	Verify a specific regeneration or erosion problem and protect occupied refuges during targeted work.	Verify the habitat function needing restoration; compare lighter measures and define stopping criteria.

Shared field requirements - check once for all options

- Vegetation regeneration and soil erosion.
- Habitat connectivity, shelter, food and breeding sites.
- Sensitive/vulnerable taxa and effects of machinery or human disturbance.

A qualified ecologist must judge whether intervention is needed and proportionate. ANWE does not replace consent or environmental permissions.

5. Design for Values Assessment

Five values guide the choice; none is reduced to an overall ethical score. This matrix describes design support and remaining evaluation, not verified ethical outcomes.

Value	What to protect	How ANWE addresses it	Current limitation
Animal Welfare	Shelter, food, movement and breeding habitat.	Considers potential animal harms in each option.	Current occupancy and site-specific impacts unknown.
Ecological Recovery	Habitat function and regeneration.	Compares lighter and more intrusive options.	Field validation of need and recovery required.
Transparency	Explainable reasoning and assumptions.	Links facts to source records and methods.	Source-backed numbers do not prove causal interpretation s.
Uncertainty	Visible weak or missing evidence.	Separates uncertain locations and missing analysis.	Sampling bias, dates, masks and July-fire effects remain.
Accountability	Human responsibility and approvals.	Records review status and justification.	Status: Unreviewed draft. Expert credentials unverified.

Value conflicts to discuss

• Rapid habitat restoration versus avoiding disturbance to wildlife.

• Useful biodiversity evidence versus protecting sensitive locations.

• Clear guidance versus honest uncertainty.

• AI convenience versus human responsibility.

Discuss these tensions with restoration organisations, land managers and field staff. Full stakeholder → value → norm → requirement → feature → evaluation traceability is in Appendix C.

6. What Happens Next?

Current decision status

Unreviewed draft. No action is authorised. Expert endorsement for further planning cannot be established because qualifications and permissions are not verified by ANWE.

Required field checks

• Survey current animal use, surviving refuges, movement routes and breeding activity.

• Check regeneration, habitat structure and erosion; determine whether lighter measures suffice.

• Record survey methods, effort and baseline monitoring indicators before comparing outcomes.

Missing environmental evidence

Validated burn-severity and current habitat/welfare evidence are unavailable. Satellite indices are available but do not fill these field gaps.

Expert Review and responsibility

A qualified expert must check evidence relevance, uncertain impacts, timing and the decision rationale. Confirm landowner consent and applicable environmental authorisations. A PDF download is not approval.

Reviewer identity and private comments are omitted by privacy settings. Public status is locally declared, not authenticated.

AI-Assisted Assessment status

An actual AI draft follows; it needs human review.

Appendix navigation: A satellite methods / B GBIF methods/species / C value traceability / D sources/licences / E glossary / F full AI draft

7. AI-Assisted Assessment

Actual model draft, not an expert-approved conclusion. Its evidence snapshot matches this report.

Editorial clarifications distinguish historical observations from current presence and remove an unsupported regeneration-rate inference. Original affected wording is retained, explicitly superseded, in Appendix F. This is not expert ecological approval.

Interpretation - excerpt

GIS and satellite data combined with biodiversity records show historical recorded observations before and after a fire event in Zamora, Spain. NDVI and dNBR indices suggest vegetation impact but are indicators, not direct damage measures. Species records from GBIF reflect observation efforts, not population counts, with birds, mammals, reptiles, amphibians, and invertebrates detected in buffer zones more than burned polygons. Restoration decisions should carefully consider animal welfare aspects such as shelter and food, alongside ecosystem recovery potential and uncertainties in data, favoring adaptive management with thorough field checks. Transparency about data limits and clear accountability in restoration actions are essential to responsibly support biodiversity post-fire.

Selected uncertainties

• Exact burned-area boundaries and severity class are not provided, limiting precise linkage between species observations and fire impacts.

• Detailed status on local soil biota and invertebrate response before and after fire is lacking due to reliance on indexed GBIF records that focus on visible fauna.

• Current occupancy or actual welfare condition of animals after the fire cannot be inferred from observation records or satellite indices alone.

Supporting evidence and assumptions

The draft selects source-owned evidence; conditional ecological implications remain hypotheses. A source reference does not prove the interpretation. Model identity, full text, all references and exact selected values are retained in Appendix F.

Human review status: Unreviewed draft. No permissions or endorsements are inferred.

Appendix A - Satellite Methodology

NDVI=(B08−B04)/(B08+B04); NBR=(B08−B12)/(B08+B12); dNBR=before NBR−later NBR. Values in the main report are rounded for readability only.

Input	Exact metadata
Analysis ID	3cfd30b240410dd72dc0b23d
Pre-image ID/date	S2B_MSIL2A_20220610T110619_N0510_R137_T29TQG_20240619T213737 2022-06-10T11:06:19.024Z
Later-image ID/date	S2A_MSIL2A_20220705T110631_N0510_R137_T29TQG_20240630T210600 2022-07-05T11:06:31.024Z
Processing grid	EPSG:3857; 60 m projected pixels; NEAREST
Spatial coverage	Regional search window; not the burned-area boundary
Created	2026-10-12T18:56:56.979347+00:00

Index	Valid pixels	Total pixels	Mean (full precision)	Range (full precision)
preNDVI	844724	892354	0.48171377182006836	Min: -0.9920397996902466 Max: 1.0
postNDVI	841197	892354	0.3849746584892273	Min: -0.9786096811294556 Max: 0.9995390176773071
dNBR	830708	892354	0.11353146284818649	Min: -0.7580866813659668 Max: 1.6078133583068848

NDVI=(B08-B04)/(B08+B04); NBR=(B08-B12)/(B08+B12); dNBR=pre NBR-post NBR; REFLECTANCE units, harmonizeValues=false; SCL/dataMask plus one-pixel cloud buffer; valid paired pixels only

SCL masking may miss cloud or smoke. Water, uncertain/dark pixels, clouds, shadows and snow are excluded; this can omit dark burned surfaces. Nearest-neighbour resampling to a common 60 m Web Mercator grid is a prototype compromise (ground spacing differs with latitude), not a native-resolution analysis. Season, drought, atmospheric correction and phenology can affect NDVI/NBR changes. Indices are not measured ecological damage, animal mortality or quantitative soil moisture. Regional statistics include unburned land unless a reviewed perimeter is applied.

Masking: SCL classes 0,1,2,3,6,7,8,9,10,11 excluded; one-pixel cloud/shadow buffer; negative/nonfinite reflectance and invalid denominators excluded; dNBR uses paired valid pixels. Dark burned surfaces can be omitted. No calibrated damage classes or wildlife mortality inference.

Public catalogue retrieval is separate from processing. A product thumbnail or tile-wide cloud percentage cannot establish valid site pixels.

Catalogue inventory: 87 retrieved products. The processing pair is documented above. Additional products referenced by an AI draft are listed below when present; unused catalogue entries are not analytical evidence.

Appendix B - GBIF Methodology and Species Lists

GBIF eventDate range search, paginated up to 2100 records per period; day-precision observation dates only. Species sets use GBIF speciesKey, not arbitrary name strings.

CEMS generalised polygons reprojected to UTM 29N (EPSG:32629); metric buffer outside burned polygon, identical geography for both windows.

Generalised display boundary; location precision unknown or uncertainty crossing a boundary is separate.

GBIF records are not a census. Counts and species lists refer to retrieved, date-eligible records, not populations. Unequal sampling effort, observer activity, reporting, seasonal variation, coverage and spatial uncertainty affect differences. Only-before does not mean disappearance; only-after does not mean recovery. No percentage-change interpretation. The June fire end date is unverified. After windows are post-start observations, not proof the incident was extinguished. The separate July 2022 wildfire may affect later records. The perimeter is a generalised CEMS observation from 25 June, not an exact final boundary.

Window	Start	End
Before	2021-06-01	2021-06-30
After-start	2023-06-01	2023-06-30

Retrieval / quality	Before	After
apiBoundingBoxCount	395	227
retrieved	395	227
truncated	False	False
missingDates	0	0
impreciseDates	0	0
outsideWindow	0	0
spatiallyUncertain	143	204
unlocated	0	0
outsideArea	204	18
geospatialIssues	0	0
excludedAbsentOrNonanimal	0	0
duplicateIds	0	0

All-date quality audit: 300 retrieved, 0 missing dates, 0 imprecise dates. Independent bounded all-date quality sample of the same bounding box; not a count of all undated records.

Taxonomic record counts

Group	Location category	Before	After
Amphibia	Inside mapped fire footprint	0	0
Amphibia	Nearby comparison area	4	0
Amphibia	Uncertain location	0	1
Aves	Inside mapped fire footprint	6	0
Aves	Nearby comparison area	6	5
Aves	Uncertain location	109	200
Insecta	Inside mapped fire footprint	3	0
Insecta	Nearby comparison area	9	0
Insecta	Uncertain location	24	2
Malacostraca	Inside mapped fire footprint	0	0
Malacostraca	Nearby comparison area	2	0
Malacostraca	Uncertain location	0	0
Mammalia	Inside mapped fire footprint	1	0
Mammalia	Nearby comparison area	9	0
Mammalia	Uncertain location	5	1
Squamata	Inside mapped fire footprint	3	0
Squamata	Nearby comparison area	5	0
Squamata	Uncertain location	5	0

Complete species sets from eligible retrieved records

All species in the retrieved sets are included. Categories describe reporting only, not disappearance or recovery. Each entry links a representative record per available window; record counts show how many eligible records support that species/category. Original dataset IDs and licences follow in Appendix D. Exact locations are withheld.

Inside mapped fire footprint

Recorded species	Window category	Records	Representative provenance
Cervus elaphus GBIF species key 2440958	Only before	1	before: gbif-3730243920 (2021-06-05)

Inside mapped fire footprint

Recorded species	Window category	Records	Representative provenance
Lullula arborea GBIF species key 2490604	Only before	2	before: gbif-3720848106 (2021-06-05)
Sylvia undata GBIF species key 2492949	Only before	2	before: gbif-3729753215 (2021-06-03)
Iphiclides feisthamelii GBIF species key 4690079	Only before	3	before: gbif-3822472122 (2021-06-04)
Chalcides striatus GBIF species key 5225191	Only before	3	before: gbif-3721379232 (2021-06-03)
Loxia curvirostra GBIF species key 9629160	Only before	2	before: gbif-3721356313 (2021-06-05)

Nearby comparison area

Recorded species	Window category	Records	Representative provenance
Lytta vesicatoria GBIF species key 1046592	Only before	1	before: gbif-3904026419 (2021-06-06)
Loweia tityrus GBIF species key 1924494	Only before	2	before: gbif-3720922577 (2021-06-04)
Lythria cruentaria GBIF species key 1959735	Only before	2	before: gbif-3905654483 (2021-06-04)
Pacifastacus leniusculus GBIF species key 2226990	Only before	2	before: gbif-3720622383 (2021-06-06)
Pelophylax perezi GBIF species key 2426658	Only before	2	before: gbif-3729159202 (2021-06-04)
Rana iberica GBIF species key 2426774	Only before	2	before: gbif-3721355088 (2021-06-04)
Cervus elaphus GBIF species key 2440958	Only before	2	before: gbif-3720676161 (2021-06-03)
Timon lepidus GBIF species key 2468542	Only before	2	before: gbif-3720357262 (2021-06-06)
Sylvia undata GBIF species key 2492949	Only before	4	before: gbif-3720807061 (2021-06-03)
Camptogramma bilineata GBIF species key 4524376	Only before	1	before: gbif-3730011174 (2021-06-04)

Nearby comparison area

Recorded species	Window category	Records	Representative provenance
Arctia tigrina GBIF species key 5115854	Only before	1	before: gbif-3731598345 (2021-06-06)
Vulpes vulpes GBIF species key 5219243	Only before	7	before: gbif-3720225135 (2021-06-04)
Zygaena trifolii GBIF species key 7761138	Only before	1	before: gbif-3904010692 (2021-06-04)
Melitaea phoebe GBIF species key 7934873	Only before	1	before: gbif-3731614335 (2021-06-04)
Malpolon monspessulanus GBIF species key 8796529	Only before	3	before: gbif-3721030364 (2021-06-04)
Loxia curvirostra GBIF species key 9629160	Only before	2	before: gbif-3721072453 (2021-06-04)
Curruca iberiae GBIF species key 10700636	Only after	1	after: gbif-4137884372 (2023-06-16)
Anthus campestris GBIF species key 2490274	Only after	2	after: gbif-4137604231 (2023-06-16)
Cuculus canorus GBIF species key 5231918	Only after	1	after: gbif-4138475710 (2023-06-16)
Alauda arvensis GBIF species key 8077224	Only after	1	after: gbif-4138056906 (2023-06-16)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Curruca iberiae GBIF species key 10700636	Both windows	6	before: gbif-3545956471 (2021-06-19) after: gbif-4761386844 (2023-06-10)
Cervus elaphus GBIF species key 2440958	Both windows	2	before: gbif-3804711467 (2021-06-04) after: gbif-4158522041 (2023-06-21)
Alectoris rufa GBIF species key 2474051	Both windows	3	before: gbif-5649321763 (2021-06-24) after: gbif-5523797170 (2023-06-25)
Merops apiaster GBIF species key 2475443	Both windows	5	before: gbif-3526500747 (2021-06-11) after: gbif-4717976661 (2023-06-10)
Alcedo atthis GBIF species key 2475532	Both windows	2	before: gbif-5480791116 (2021-06-17) after: gbif-4658537938 (2023-06-28)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Gyps fulvus GBIF species key 2480389	Both windows	3	before: gbif-3532881335 (2021-06-30) after: gbif-4657577365 (2023-06-10)
Ciconia ciconia GBIF species key 2481912	Both windows	5	before: gbif-5599862711 (2021-06-17) after: gbif-5487524512 (2023-06-22)
Cyanistes caeruleus GBIF species key 2487879	Both windows	5	before: gbif-3527588863 (2021-06-30) after: gbif-4661582450 (2023-06-10)
Oriolus oriolus GBIF species key 2488949	Both windows	9	before: gbif-5553553315 (2021-06-07) after: gbif-4655620633 (2023-06-10)
Sturnus unicolor GBIF species key 2489104	Both windows	5	before: gbif-5439505199 (2021-06-24) after: gbif-5582714322 (2023-06-25)
Delichon urbicum GBIF species key 2489214	Both windows	5	before: gbif-5483720367 (2021-06-02) after: gbif-4753049573 (2023-06-08)
Lullula arborea GBIF species key 2490604	Both windows	4	before: gbif-5598751595 (2021-06-18) after: gbif-5423523486 (2023-06-25)
Turdus merula GBIF species key 2490719	Both windows	7	before: gbif-5456939619 (2021-06-18) after: gbif-4635232340 (2023-06-08)
Turdus viscivorus GBIF species key 2490774	Both windows	3	before: gbif-5578403675 (2021-06-18) after: gbif-5548175853 (2023-06-13)
Sylvia undata GBIF species key 2492949	Both windows	3	before: gbif-3528928206 (2021-06-19) after: gbif-4754572887 (2023-06-10)
Sylvia atricapilla GBIF species key 2492956	Both windows	3	before: gbif-5484671383 (2021-06-17) after: gbif-4657454726 (2023-06-10)
Phylloscopus bonelli GBIF species key 2493070	Both windows	5	before: gbif-3527772598 (2021-06-30) after: gbif-4752456068 (2023-06-10)
Serinus serinus GBIF species key 2494200	Both windows	5	before: gbif-5419984254 (2021-06-02) after: gbif-4633880482 (2023-06-10)
Fringilla coelebs GBIF species key 2494422	Both windows	13	before: gbif-5332930077 (2021-06-02) after: gbif-4663171014 (2023-06-10)
Carduelis carduelis GBIF species key 2494686	Both windows	4	before: gbif-5630579452 (2021-06-07) after: gbif-5452776722 (2023-06-14)
Columba livia GBIF species key 2495414	Both windows	2	before: gbif-5515032574 (2021-06-02) after: gbif-5470004673 (2023-06-03)
Columba palumbus GBIF species key 2495455	Both windows	9	before: gbif-5592057719 (2021-06-24) after: gbif-4685706431 (2023-06-10)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Saxicola rubicola GBIF species key 4408759	Both windows	12	before: gbif-3523435395 (2021-06-11) after: gbif-4753858561 (2023-06-10)
Ptyonoprogne rupestris GBIF species key 4408796	Both windows	3	before: gbif-5529666073 (2021-06-24) after: gbif-4655188263 (2023-06-08)
Apus apus GBIF species key 5228676	Both windows	8	before: gbif-5338012300 (2021-06-24) after: gbif-5361105263 (2023-06-03)
Milvus migrans GBIF species key 5229167	Both windows	14	before: gbif-3522586118 (2021-06-11) after: gbif-4670550957 (2023-06-10)
Milvus milvus GBIF species key 5229168	Both windows	4	before: gbif-3530886157 (2021-06-30) after: gbif-4838790052 (2023-06-10)
Pica pica GBIF species key 5229490	Both windows	10	before: gbif-3528973938 (2021-06-30) after: gbif-5356868105 (2023-06-25)
Garrulus glandarius GBIF species key 5229493	Both windows	2	before: gbif-5497256765 (2021-06-24) after: gbif-5333798617 (2023-06-02)
Passer domesticus GBIF species key 5231190	Both windows	8	before: gbif-5309297450 (2021-06-18) after: gbif-4832861535 (2023-06-10)
Cuculus canorus GBIF species key 5231918	Both windows	6	before: gbif-5541361843 (2021-06-17) after: gbif-4689342161 (2023-06-10)
Phoenicurus ochruros GBIF species key 5739315	Both windows	6	before: gbif-5346432120 (2021-06-02) after: gbif-4761041058 (2023-06-10)
Chroicocephalus ridibundus GBIF species key 6065824	Both windows	2	before: gbif-5683238704 (2021-06-24) after: gbif-4762646273 (2023-06-01)
Coloeus monedula GBIF species key 6100954	Both windows	2	before: gbif-5477301389 (2021-06-24) after: gbif-4791933089 (2023-06-08)
Lanius meridionalis GBIF species key 7341500	Both windows	4	before: gbif-5473781444 (2021-06-24) after: gbif-4641347447 (2023-06-10)
Luscinia megarhynchos GBIF species key 7660710	Both windows	10	before: gbif-5373081855 (2021-06-17) after: gbif-4797139487 (2023-06-08)
Lanius collurio GBIF species key 7745240	Both windows	4	before: gbif-5370014787 (2021-06-17) after: gbif-5465233456 (2023-06-21)
Picus sharpei GBIF species key 9029556	Both windows	8	before: gbif-5446100825 (2021-06-24) after: gbif-4713932234 (2023-06-10)
Corvus corone GBIF species key 9409796	Both windows	5	before: gbif-3534591650 (2021-06-11) after: gbif-5495758293 (2023-06-25)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Falco tinnunculus GBIF species key 9616058	Both windows	5	before: gbif-3624370425 (2021-06-11) after: gbif-4761321479 (2023-06-10)
Motacilla alba GBIF species key 9687165	Both windows	6	before: gbif-5410553637 (2021-06-17) after: gbif-5540457495 (2023-06-22)
Parus major GBIF species key 9705453	Both windows	3	before: gbif-5646868893 (2021-06-18) after: gbif-4838490669 (2023-06-10)
Lytta vesicatoria GBIF species key 1046592	Only before	2	before: gbif-4931403357 (2021-06-06)
Boyeria irene GBIF species key 1425137	Only before	1	before: gbif-3408123466 (2021-06-27)
Loweia tityrus GBIF species key 1924494	Only before	1	before: gbif-4941693006 (2021-06-04)
Leptotes pirithous GBIF species key 1932976	Only before	3	before: gbif-3720985493 (2021-06-04)
Lythria cruentaria GBIF species key 1959735	Only before	2	before: gbif-4881169425 (2021-06-04)
Talpa occidentalis GBIF species key 2436152	Only before	2	before: gbif-3729663081 (2021-06-04)
Timon lepidus GBIF species key 2468542	Only before	4	before: gbif-3720824976 (2021-06-06)
Corvus corax GBIF species key 2482492	Only before	2	before: gbif-3529069206 (2021-06-11)
Petronia petronia GBIF species key 2492350	Only before	1	before: gbif-5676207058 (2021-06-24)
Camptogramma bilineata GBIF species key 4524376	Only before	1	before: gbif-4881480257 (2021-06-04)
Euphydryas aurinia GBIF species key 4535809	Only before	4	before: gbif-3720767695 (2021-06-04)
Iphiclides feisthamelii GBIF species key 4690079	Only before	1	before: gbif-4941693005 (2021-06-04)
Arctia tigrina GBIF species key 5115854	Only before	2	before: gbif-4883516189 (2021-06-06)
Vulpes vulpes GBIF species key 5219243	Only before	2	before: gbif-3720937179 (2021-06-03)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Prunella modularis GBIF species key 5231763	Only before	2	before: gbif-3539466972 (2021-06-19)
Chloris chloris GBIF species key 5845582	Only before	2	before: gbif-5611419659 (2021-06-07)
Emberiza cirlus GBIF species key 7626513	Only before	2	before: gbif-5453775627 (2021-06-18)
Zygaena trifolii GBIF species key 7761138	Only before	4	before: gbif-4880792810 (2021-06-04)
Melitaea phoebe GBIF species key 7934873	Only before	3	before: gbif-4888898782 (2021-06-04)
Malpolon monspessulanus GBIF species key 8796529	Only before	1	before: gbif-3720489569 (2021-06-02)
Issoria lathonia GBIF species key 1905383	Only after	2	after: gbif-4137950988 (2023-06-16)
Rana iberica GBIF species key 2426774	Only after	1	after: gbif-4137939199 (2023-06-17)
Dendrocopos major GBIF species key 2477968	Only after	2	after: gbif-4648262273 (2023-06-10)
Pernis apivorus GBIF species key 2480420	Only after	1	after: gbif-4677549766 (2023-06-10)
Accipiter nisus GBIF species key 2480637	Only after	1	after: gbif-4834955851 (2023-06-12)
Hieraaetus pennatus GBIF species key 2480685	Only after	3	after: gbif-4613011076 (2023-06-10)
Egretta garzetta GBIF species key 2480876	Only after	1	after: gbif-4720517948 (2023-06-08)
Falco subbuteo GBIF species key 2481035	Only after	1	after: gbif-4685236786 (2023-06-28)
Chlidonias niger GBIF species key 2481119	Only after	1	after: gbif-4763349934 (2023-06-01)
Phalacrocorax carbo GBIF species key 2481890	Only after	4	after: gbif-4641297193 (2023-06-28)
Podiceps cristatus GBIF species key 2482054	Only after	1	after: gbif-4838734979 (2023-06-08)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Periparus ater GBIF species key 2487871	Only after	1	after: gbif-5484120267 (2023-06-27)
Galerida cristata GBIF species key 2490669	Only after	1	after: gbif-4838974999 (2023-06-10)
Emberiza hortulana GBIF species key 2491482	Only after	2	after: gbif-5315091962 (2023-06-25)
Emberiza cia GBIF species key 2491498	Only after	1	after: gbif-4137705757 (2023-06-16)
Erithacus rubecula GBIF species key 2492462	Only after	2	after: gbif-5356018720 (2023-06-02)
Lanius senator GBIF species key 2492867	Only after	3	after: gbif-4640631764 (2023-06-10)
Streptopelia decaocto GBIF species key 2495696	Only after	1	after: gbif-5657734508 (2023-06-02)
Streptopelia turtur GBIF species key 2495708	Only after	3	after: gbif-4756419637 (2023-06-08)
Clamator glandarius GBIF species key 2496464	Only after	2	after: gbif-5487530910 (2023-06-25)
Athene noctua GBIF species key 2497266	Only after	2	after: gbif-5393183006 (2023-06-25)
Otus scops GBIF species key 2497701	Only after	2	after: gbif-5437994877 (2023-06-25)
Upupa epops GBIF species key 2498415	Only after	3	after: gbif-4774298901 (2023-06-10)
Aegypius monachus GBIF species key 5229165	Only after	3	after: gbif-4787642563 (2023-06-10)
Oenanthe hispanica GBIF species key 5231235	Only after	1	after: gbif-5514772794 (2023-06-13)
Oenanthe oenanthe GBIF species key 5231240	Only after	2	after: gbif-5529744742 (2023-06-25)
Troglodytes troglodytes GBIF species key 5231438	Only after	5	after: gbif-4612980077 (2023-06-10)
Phylloscopus ibericus GBIF species key 7341585	Only after	1	after: gbif-4662683225 (2023-06-08)

Uncertain location

Recorded species	Window category	Records	Representative provenance
Emberiza calandra GBIF species key 7634625	Only after	4	after: gbif-4748213627 (2023-06-10)
Certhia brachydactyla GBIF species key 7806848	Only after	2	after: gbif-5396973263 (2023-06-25)
Alauda arvensis GBIF species key 8077224	Only after	2	after: gbif-5432486185 (2023-06-25)
Linaria cannabina GBIF species key 8104397	Only after	1	after: gbif-4766558248 (2023-06-10)
Caprimulgus europaeus GBIF species key 8109681	Only after	2	after: gbif-5419525236 (2023-06-25)
Hirundo rustica GBIF species key 9515886	Only after	3	after: gbif-4835756107 (2023-06-10)
Anas platyrhynchos GBIF species key 9761484	Only after	4	after: gbif-4633728523 (2023-06-08)
Ardea cinerea GBIF species key 9797180	Only after	2	after: gbif-4713931991 (2023-06-28)
Curruca melanocephala GBIF species key 9990095	Only after	1	after: gbif-4773633562 (2023-06-10)

Appendix C - Design for Values Traceability

Stakeholder concern → Value → Norm → Requirement → Feature → Evaluation. The chain describes how the application was designed, not verified ecological outcomes.

Animal welfare

Chain element	Project specification
Stakeholder concern	Restoration organisations and field staff may unintentionally disturb shelter, feeding, movement or breeding.
Norm	Avoid unnecessary welfare harms
Requirement	Show likely effects on shelter, food, movement and breeding habitat
Feature	Four explicit welfare dimensions in each pathway and AI assessment
Evaluation	Check all four dimensions are assessed as conditional hypotheses or explicitly unknown; verify with field surveys.

Ecological recovery

Chain element	Project specification
Stakeholder concern	Land managers need locally suitable regeneration, habitat structure and erosion responses.
Norm	Support locally appropriate recovery
Requirement	Compare effects on habitat structure and regeneration
Feature	Three restoration options, spectral evidence and field checks
Evaluation	Check vegetation, structure, regeneration and soil/erosion needs; do not interpret NDVI as ecosystem recovery by itself.

Transparency

Chain element	Project specification
Stakeholder concern	Ecologists, agencies and funders cannot independently evaluate unsupported explanations.
Norm	Recommendations must be explainable
Requirement	Show supporting evidence and assumptions
Feature	Source-linked factual panels, structured evidence fields and report methodology
Evaluation	Trace each factual AI item to a server-owned source field; inspect methods, assumptions and licences.

Uncertainty

Chain element	Project specification
Stakeholder concern	Landowners need usable guidance while researchers need sampling biases and unknown impacts disclosed.
Norm	Weak evidence must remain visible
Requirement	Display evidence quality and flag the need for field validation
Feature	Unknown labels, GBIF coverage warnings, uncertainty and survey requirements
Evaluation	Check incomplete sampling, dates, spatial uncertainty and absent evidence are flagged; no absence or recovery claims from occurrence counts.

Accountability

Chain element	Project specification
Stakeholder concern	Organisations, managers and public agencies need a clear responsibility and approval trail.
Norm	Humans remain responsible
Requirement	Require expert approval before action
Feature	Explicit review status, decision justification and approval checklist; no execution capability
Evaluation	Block a locally recorded reviewed decision unless reviewer, rationale, preferred option and confirmation exist; always retain external approval requirements.

Stakeholder priorities

• Primary - Restoration organisations and ecologists: Traceable evidence, animal welfare checks and defensible recovery options.

• Primary - Landowners and land managers: Feasible decisions, timing, habitat function and explicit responsibilities.

• Primary - Field staff and volunteers: Understandable survey requirements, disturbance precautions and escalation.

• Secondary - Funders and donors: Transparent progress and justified spending without oversimplified outcome scores.

• Secondary - Public environmental agencies: Methodology, approval requirements and accountable human decisions.

• Secondary - Researchers and ecological data providers: Attribution, data quality, privacy and reproducible comparisons.

Stakeholder needs are authored project assumptions. Empirical stakeholder and ecological outcome evaluation remain pending.

Appendix D - Data Sources, Licences and Methodology

CEMS provides the mapped extent; Copernicus Data Space supplies Sentinel imagery/metadata; GBIF indexes records from contributing datasets. No historical EFFIS severity raster or perimeter was retrieved.

CEMS activation EMSR580

EFFIS - access limitation, not a dataset used

ANWE processing of CDSE Sentinel-2 L2A reflectance

Dataset/source ID: 3cfd30b240410dd72dc0b23d | Observation/retrieval date: 2022-07-05T11:06:31.024Z

https://documentation.dataspace.copernicus.eu/APIs/SentinelHub/Data/S2L2A.html

Method: NDVI=(B08-B04)/(B08+B04); NBR=(B08-B12)/(B08+B12); dNBR=pre NBR-post NBR; REFLECTANCE units, harmonizeValues=false; SCL/dataMask plus one-pixel cloud buffer; valid paired pixels only

Limitations: SCL masking may miss cloud or smoke. Water, uncertain/dark pixels, clouds, shadows and snow are excluded; this can omit dark burned surfaces. Nearest-neighbour resampling to a common 60 m Web Mercator grid is a prototype compromise (ground spacing differs with latitude), not a native-resolution analysis. Season, drought, atmospheric correction and phenology can affect NDVI/NBR changes. Indices are not measured ecological damage, animal mortality or quantitative soil moisture. Regional statistics include unburned land unless a reviewed perimeter is applied.

Reuse: Copernicus Sentinel legal notice (collection licence: other)

Copernicus EMS Rapid Mapping

Dataset/source ID: EMSR580 | Observation/retrieval date: 2022-06-15

https://mapping.emergency.copernicus.eu/activations/EMSR580/

Method: Documented activation narrative; not a field measurement or final perimeter

Limitations: Activation documents fires starting 15 June. No extinguishment date inferred. June and July 2022 fires must not be conflated. Search-window records are not necessarily in burned habitat.

Reuse: Copernicus EMS Terms and Conditions

Copernicus EMS Rapid Mapping

Dataset/source ID: EMSR580_AOI01_GRA_PRODUCT_r1_RTP01_v1 | Observation/retrieval date: 2022-06-25T10:44:58Z

https://mapping.emergency.copernicus.eu/activations/EMSR580/

Method: 505 original mapped burnt-area polygons; topology-preserving simplification at 0.00015 degrees for display. Original source retained locally; no invented outline or area estimate.

Limitations: Mapped burnt area observed on 25 June, not necessarily final incident extent. Display generalisation is not suitable for cadastral or pixel-level decisions. No EFFIS attribution or severity class inferred.

Reuse: Copernicus EMS Terms and Conditions; see source metadata for third-party rights

ANWE temporal comparison of GBIF indexed observations

Dataset/source ID: aade3a4b643cb310 | Observation/retrieval date: 2026-10-12T18:20:04.646006+00:00

https://techdocs.gbif.org/en/openapi/v1/occurrence

Method: GBIF eventDate range search, paginated up to 2100 records per period; day-precision observation dates only. Species sets use GBIF speciesKey, not arbitrary name strings.

Limitations: GBIF records are not a census. Counts and species lists refer to retrieved, date-eligible records, not populations. Unequal sampling effort, observer activity, reporting, seasonal variation, coverage and spatial uncertainty affect differences. Only-before does not mean disappearance; only-after does not mean recovery. No percentage-change interpretation. The June fire end date is unverified. After windows are post-start observations, not proof the incident was extinguished. The separate July 2022 wildfire may affect later records. The perimeter is a generalised CEMS observation

from 25 June, not an exact final boundary. Retrieved subsets are insufficient for population trends or causal fire effects.Missing coordinate precision is reported separately. Species names in model context capped at thirty per zone/period.

Reuse: GBIF constituent dataset licences retained in comparison records

GBIF occurrence index

Dataset/source ID: 50c9509d-22c7-4a22-a47d-8c48425ef4a7 | Observation/retrieval date: 2026-03-16T08:41:47

https://www.gbif.org/occurrence/6179095254

Method: GBIF georeferenced Animalia search; indexed observation, not independently verified occupancy

Limitations: Historical or recent record does not prove current occupancy or fire exposure. Sampling is incomplete and biased; missing records do not imply absence. Exact coordinates intentionally withheld. Native status and conservation status not verified.

Reuse: http://creativecommons.org/licenses/by-nc/4.0/legalcode

GBIF occurrence index

Dataset/source ID: 8a863029-f435-446a-821e-275f4f641165 | Observation/retrieval date: 2026-01-25

https://www.gbif.org/occurrence/6182790201

Method: GBIF georeferenced Animalia search; indexed observation, not independently verified occupancy

Limitations: Historical or recent record does not prove current occupancy or fire exposure. Sampling is incomplete and biased; missing records do not imply absence. Exact coordinates intentionally withheld. Native status and conservation status not verified.

Reuse: http://creativecommons.org/licenses/by-nc/4.0/legalcode

GBIF contributing datasets

Dataset ID	Provider / reuse information
50c9509d-22c7-4a22-a47d-8c48425ef4a7	iNaturalist research-grade observations http://creativecommons.org/licenses/by-nc/4.0/legalcode
4fa7b334-ce0d-4e88-aaae-2e0c138d049e	Dataset 4fa7b334-ce0d-4e88-aaae-2e0c138d049e http://creativecommons.org/licenses/by/4.0/legalcode
8a863029-f435-446a-821e-275f4f641165	Dataset 8a863029-f435-446a-821e-275f4f641165 http://creativecommons.org/licenses/by-nc/4.0/legalcode
3ccfe992-043a-4fff-951d-1e6e158854da	GBIF MammalNet http://creativecommons.org/licenses/by/4.0/legalcode
73319645-c321-48ed-b923-b3782a203589	Distribution database of arthropods of the Iberian Peninsula and Balearic Islands http://creativecommons.org/licenses/by-nc/4.0/legalcode

Appendix E - How to Read This Report: Glossary

Term	Plain-language meaning
GBIF	Global Biodiversity Information Facility. A network and online index of biodiversity records contributed by many organisations. ANWE uses it to find historical animal records, not to count all animals currently living in the area.
Occurrence record	A record that a taxon was reported at a place and time. It may come from an observation or collected specimen. Many records do not necessarily mean many animals; none does not prove absence.
Distinct species	Different species identified in the eligible records, counting each species once. For example, ten records of the same species still count as one distinct species. This is not the total number of species living there.
Taxon / taxonomic group	A biological classification, such as a species or class. In the tables, Aves means birds, Mammalia mammals, Amphibia amphibians, Insecta insects, Arachnida spiders and relatives, Squamata lizards and snakes, and Malacostraca a group of crustaceans.
NDVI	Normalized Difference Vegetation Index. A satellite measure of vegetation greenness using red and near-infrared light. Values usually range from -1 to 1; higher values often indicate more green vegetation. It is not a direct measure of habitat quality, recovery or animal wellbeing.
NBR	Normalized Burn Ratio. A satellite index using near-infrared and short-wave infrared light to investigate burned land and vegetation change. ANWE calculates it for the selected before/after images; interpretation requires context.
dNBR	Differenced Normalized Burn Ratio: before-image NBR minus after-image NBR. A positive value means NBR was lower in the later image, which can be associated with fire-related change. ANWE does not turn this into validated damage classes or animal deaths.
Index versus percentage	An index is a calculated indicator, not a percentage. A dNBR value of 0.1 does not mean ten percent damage. Differences can also reflect drought, season, clouds and other influences.
Sentinel-2 / Copernicus	Sentinel-2 is a satellite mission in Europe's Copernicus Earth-observation programme. Its images provide the light measurements used for ANWE's vegetation indices. A full-tile preview is an overview image, not a calculated index map.
Level-2A / surface reflectance	Satellite imagery processed to estimate light reflected by the ground after atmospheric correction. This supports index calculations but does not remove every cloud, smoke or measurement problem.
CEMS / EMSR580	Copernicus Emergency Management Service supplies disaster mapping. EMSR580 identifies the mapping activation used for this wildfire. ANWE displays its mapped burned extent observed on 25 June 2022, generalised for display.
EFFIS	European Forest Fire Information System. A source of European wildfire information. Historical June perimeter and severity data were not retrieved for this demo; the CEMS outline is a separate source.

Term	Plain-language meaning
Burned perimeter / extent	The mapped fire footprint: land mapped as burned on 25 June 2022. The same outline classifies records in both periods. Before-fire records refer to places later mapped as burned, not animals known to have experienced the fire. The outline is generalised, not an exact final boundary.
Buffer	Nearby comparison land outside the mapped fire footprint, extending the selected distance outward from its boundary. A five-kilometre buffer is not a radius from the centre. This land is not necessarily unaffected by fire or a validated experimental control.
Spatial uncertainty	The reported point is within the selected study footprint, but location precision is unknown or could cross a boundary. Such records are not confidently assigned to the inside/nearby categories. Uncertain is a data-quality category, not a third physical area or a damage class.
Pixel / raster / masked pixel	A raster is a grid of image cells called pixels. Each cell represents ground area. Masked pixels are excluded because of cloud, shadow, water, missing data or quality checks; a blank pixel is not evidence of no damage or no animals.
Mean / valid pixels	The mean is the arithmetic average of usable values. Valid pixels are the cells that passed the analysis checks. ANWE's regional mean includes unburned land and does not describe every location or only the fire perimeter.
Observed / Derived / Hypothetical / Unknown	Observed: retrieved source records. Derived: results calculated using a stated method. Hypothetical: a plausible possibility requiring verification. Unknown: no adequate supporting information. These labels describe evidence status, not certainty that a conclusion is correct.
Design for Values / DfV	The ethical framework connecting stakeholder concerns to values, norms, design requirements, features and evaluation. ANWE considers Animal welfare, Ecological recovery, Transparency, Uncertainty and Accountability separately, without one overall ethical score.
Field verification / expert review	Field verification means checking conditions on the ground, such as animal use, refuges, breeding activity and erosion. Expert review means a qualified person evaluates the evidence and decision. A downloadable draft or AI answer is not permission to implement restoration.

https://techdocs.gbif.org/en/data-publishing/dataset-classes

https://www.usgs.gov/landsat-missions/landsat-normalized-difference-vegetation-index

https://www.usgs.gov/landsat-missions/landsat-normalized-burn-ratio

Appendix F - Full AI Draft and Evidence References

Actual model: gpt-4.1-mini-2025-04-14; generated 2026-10-12T20:30:26.453844+00:00

Question: Which species were recorded before and after the fire, what do the observation counts and satellite indices mean, and how do the five ANWE values affect restoration? Discuss possible shelter, food, water, movement and breeding concerns for mammals, birds, reptiles, amphibians, invertebrates and soil biota; distinguish evidence from hypotheses and specify field checks. Do not infer current occupancy or measured welfare.

Evidence snapshot: 123f2ed1cf053a4f

Unvalidated AI draft. Earlier selected drafts are superseded by this latest assessment; no AI is generated during export.

Editorial corrections - superseded original wording

The following original model passages are retained for traceability and must not be treated as supported findings. The displayed draft incorporates the clarifications. Other AI interpretations remain unreviewed.

Original model wording (superseded)	Editorial clarification
varied species presence	historical recorded observations
Vegetation indices show decline consistent with fire impact, likely slowing habitat regeneration.	The regional indices differ between the selected images. They do not establish the cause of change or the rate of habitat regeneration.

GIS and satellite data combined with biodiversity records show historical recorded observations before and after a fire event in Zamora, Spain. NDVI and dNBR indices suggest vegetation impact but are indicators, not direct damage measures. Species records from GBIF reflect observation efforts, not population counts, with birds, mammals, reptiles, amphibians, and invertebrates detected in buffer zones more than burned polygons. Restoration decisions should carefully consider animal welfare aspects such as shelter and food, alongside ecosystem recovery potential and uncertainties in data, favoring adaptive management with thorough field checks. Transparency about data limits and clear accountability in restoration actions are essential to responsibly support biodiversity post-fire.

Evidence ID	Source field	Exact selected value
comparison-aade3a4b643cb310	before.burned.distinctSpecies	6
comparison-aade3a4b643cb310	after.buffer.distinctSpecies	4
analysis-3cfd30b240410dd72dc0b23d	preNDVI.mean	0.48171377182006836
analysis-3cfd30b240410dd72dc0b23d	postNDVI.mean	0.3849746584892273
analysis-3cfd30b240410dd72dc0b23d	dNBR.mean	0.11353146284818649
cems-EMSR580-event	eventStart	2022-06-15

Hypotheses

• Observation counts from GBIF (Global Biodiversity Information Facility) represent recorded instances of species occurrences within searched regions and time windows, affected by variable sampling effort and dates, not actual population sizes or proven presence at fire-affected sites.

• Changes in NDVI (Normalized Difference Vegetation Index) and dNBR (differenced Normalized Burn Ratio) satellite indices indicate general changes in vegetation greenness and burned area reflectance, not exact measures of ecological damage or animal welfare.

• Post-fire availability of shelter, food, water, movement corridors, and breeding locations likely varies by species group and habitat condition, influencing recovery potential and welfare.

• Restoration values such as Animal welfare and Ecological recovery must consider habitat structure, food chains, soil condition, and ecosystem connectivity, balancing intervention benefits and disturbances.

Unknowns

• Exact burned-area boundaries and severity class are not provided, limiting precise linkage between species observations and fire impacts.

• Detailed status on local soil biota and invertebrate response before and after fire is lacking due to reliance on indexed GBIF records that focus on visible fauna.

• Current occupancy or actual welfare condition of animals after the fire cannot be inferred from observation records or satellite indices alone.

• Water availability changes and small-scale habitat microclimates affecting amphibians and reptiles are not directly assessed here.

Animal welfare

Fire may have reduced shelter and food availability temporarily, affecting mammals, birds, reptiles, amphibians, invertebrates, and soil organisms which require diverse microhabitats and resources for movement and breeding. Welfare is sensitive to habitat structure changes, which should be monitored carefully.

• Field surveys examining availability of shelters like logs, burrows, vegetation cover and food sources immediately post-fire and over time.

Evidence IDs: comparison-aade3a4b643cb310

Ecological recovery

The regional indices differ between the selected images. They do not establish the cause of change or the rate of habitat regeneration. Recovery of diverse species groups depends on soil condition, seed sources, and connectivity; progress should be tracked via repeated remote sensing and species monitoring.

• Repeated NDVI, dNBR monitoring combined with species observations to assess regeneration and habitat quality.

Evidence IDs: analysis-3cfd30b240410dd72dc0b23d

Transparency

Data sources openly document species observations, fire event timing, and satellite assessments, supporting informed discourse and decision-making.

• Accessible reporting of raw and processed data, methods, and limitations.

Evidence IDs: cems-EMSR580-event, comparison-aade3a4b643cb310

Uncertainty

Observations and satellite indices have limits, including spatial and temporal uncertainties, varied sampling efforts, and lack of fire severity detail, cautioning against direct inference of species absence or recovery.

• Cross-validation of indices with field surveys and expert ecological assessment.

Evidence IDs: analysis-3cfd30b240410dd72dc0b23d, comparison-aade3a4b643cb310

Accountability

Responsible management requires recognizing data gaps, avoiding assumptions about populations, and ensuring restoration actions respect species needs and ecosystem complexity.

• Clear documentation of restoration goals, monitoring plans, and involvement of ecological experts.

Evidence IDs: comparison-aade3a4b643cb310

Natural regeneration and monitoring

Potential benefits: Allows ecosystems to recover with minimal human disturbance, preserving natural succession processes that support shelter and food availability for diverse species.

Possible harms: Recovery may be slow; species sensitive to fire damage or with specific habitat needs might decline without support.

When inappropriate: When critical habitats or keystone species are severely impacted, purely passive recovery may fail to sustain biodiversity.

Field checks: Repeat surveys to monitor species presence and habitat regeneration across burned and buffer areas to detect natural recovery rates and risks.

Evidence IDs: analysis-3cfd30b240410dd72dc0b23d, comparison-aade3a4b643cb310

Assisted ecological restoration

Potential benefits: Targeted actions (e.g., habitat structure improvement, erosion control) can improve food sources, shelter, and breeding sites important for mammals, birds, reptiles, amphibians, soil organisms, and invertebrates.

Possible harms: Intervention disturbance may stress wildlife, disrupt breeding, or introduce invasive species if poorly planned.

When inappropriate: When natural regeneration is sufficient or when interventions risk further harm to sensitive species or soil biota.

Field checks: Evaluation of soil condition, vegetation cover quality, and species habitat use before and after intervention to measure effectiveness and avoid harm.

Evidence IDs: analysis-3cfd30b240410dd72dc0b23d

Targeted habitat restoration

Potential benefits: Focused restoration of critical habitats can facilitate movement corridors, secure breeding grounds, and preserve food webs essential to ecological recovery.

Possible harms: May require substantial resources and risk unintended ecological changes if species needs and interactions are not well understood.

When inappropriate: In heterogeneous or large burned areas where detailed data is lacking or conditions are rapidly changing.

Field checks: Detailed habitat mapping and species behavioral studies to tailor restoration to specific ecological requirements and minimize adverse effects.

Evidence IDs: cems-EMSR580-mapped-extent